BEHAVIORAL BASIS FOR MOMENTUM

Fama (1998) suggested that behavioral biases could be subject to "model dredging," where one tries to find biases to fit the facts. He speculated that there might be an increasing number of behavioral models attempting to explain momentum profits. This never happened. In fact, the half dozen or so behavioral explanations for momentum that existed at that time are the same ones that have continued up to the present. Instead, as we just saw, it was the supporters of efficient markets who kept trying to find additional risk factors they hoped would explain momentum on a rational basis. These attempts were similar to the way researchers continued to look for additional risk factors to shore up the problem-prone linear factor models described in Chapter 3.

EARLY BEHAVIORAL MODELING

Social psychology has always had a strong connection with stock market investing. In 1912, Selden wrote *Psychology of the Stock Market*, based on "the belief that the movements of prices on the exchanges are dependent to a very considerable degree on the mental attitude of the investing and trading public." According to Graham and Dodd (1951), "The prices of common stocks are not carefully thought out computations, but the resultants of a welter of human reactions."

The most cited paper to ever appear in the prestigious economics journal *Econometrica* was "Prospect Theory: An Analysis of Decision Under Risk," by two psychologists, Kahneman and Tversky (1979). Kahneman received the Nobel Prize in Economic Sciences in 2002 and the Presidential Medal of Freedom in 2013. (Tversky had passed away earlier.) These authors' groundbreaking paper challenged conventional utility-maximizing behavior. Prospect theory showed that people value gains differently than they value losses. Investors being more sensitive to losses than they are to gains became known as "loss aversion." Prospect theory helped explain why individuals make decisions that can deviate from rational decision making.

The work of Kahneman and Tversky formed the basis for identifying other systematic behavioral biases that express themselves in the following